IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

EEE PART III/IV: MEng, BEng and ACGI

Corrected copy

ELECTRICAL ENERGY SYSTEMS

Thursday, 6 December 9:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer 2 questions from Section A and 2 questions from Section B. Use a separate answer book for each section.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri, F. Teng

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Section A – Answer any 2 out of 3 questions in section A

Question 1

a) What are the key objectives for operation and design of electrical energy systems?

[3]

b) Write the time expressions of currents and voltages for a three-phase system and derive the expressions for three phase active power and reactive power.

[6]

c) Measurement demonstrates that diversified peak demand of 100 households is 300kW, while the peak demand of each individual household is 6kW.

(i) Calculate the coincidence factor of an infinite number of household.

[3]

(ii) Briefly explain the benefits of diversification.

[1]

d) There are three identical generators, each of 90 kW capacity and availability of 98%. Assuming the reliability criterion requires that the probability to meet peak demand is above 95%, how many households [as in c)] can be supplied by these 3 generators?

[7]

Question 2

a) What are the advantages to apply per-unit system in the analysis of power systems?

[3]

b) Consider a high voltage transmission circuit in Figure 1. Assume that voltages (magnitudes and phase angles) are known at both sending and receiving ends.

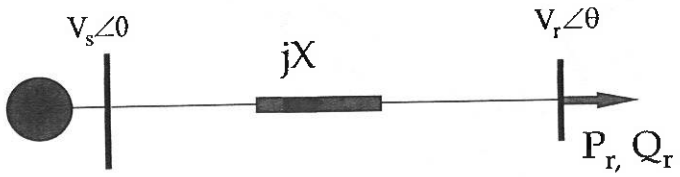


Figure 1 Transmission circuit with voltages specified at the both sending and receiving ends

(i) Write down the expressions for sending end active and reactive powers as functions of voltage magnitudes and phase angles.

[4]

(ii) Based on these expressions, show that active power flow requires a difference in phase angle and that reactive power flow requires a difference in voltage magnitude.

[2]

(iii) What is the maximum amount of active power that can be transported via this transmission line?

[2]

c) There is a three-phase system as show in Figure 2. A generator is producing 45MW with 0.5PF lagging at15kV, which is fed into a 11kV substation. Please calculate the voltage (Vr) and the complex power (Sr) at the receiving end.

[9]

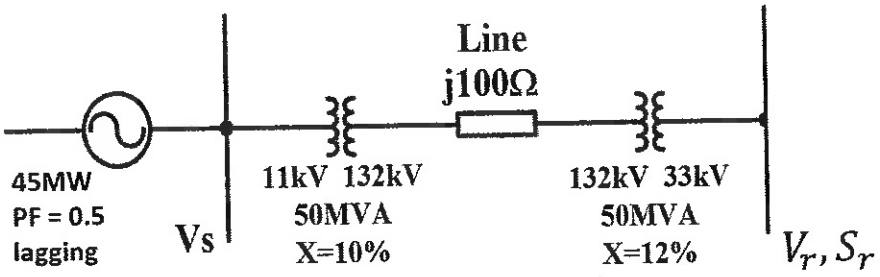


Figure 2 Three-phase system is supplied by a generator running at 45MW with 0.5PF lagging

Question 3

A three-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per unit values.

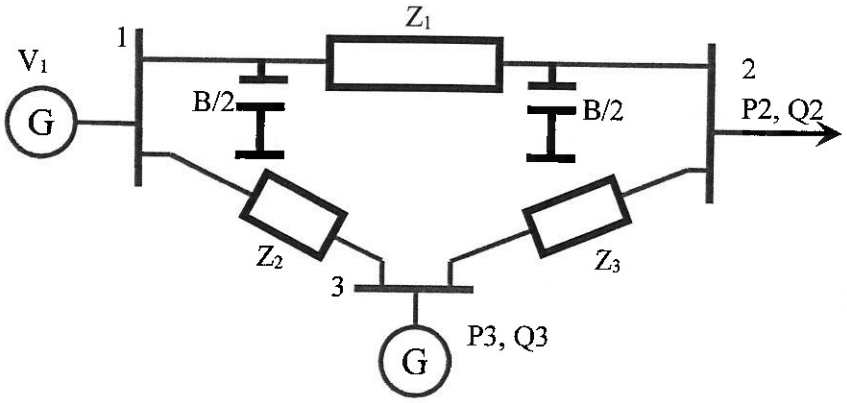


Figure 3. A three-bus power network

a) Explain the concept of the slack bus, PV bus and PQ bus in the load flow calculations

[5]

b) Given Z_1, Z_2, Z_3 and B, form the Ybus matrix

[6]

c) Assuming the per unit values $|V_1| = 1$, $|V_3| = 1.05$, $P_2 = 1$, $Q_2 = 1$, $P_3 = 1$, $Z_1 = 0.2j$, $Z_2 = 0.25j$, $Z_3 = 0.25j$ and $B = 0$, perform 2 iterations of the Gauss-Seidel load flow method.

[9]

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Section B – Answer any 2 out of 3 questions in section B

Question 4

a) The positive sequence sub-transient reactance of a synchronous generator is 0.10 p.u. while its positive sequence steady-state reactance is 0.01 p.u. Justify or rectify the statement with proper explanation.

[4]

b) State the basic similarities and differences between power flow and short-circuit analyses in terms of the following:

(i) Type of the equations to be solved.

[2]

(ii) Solution technique used.

[2]

(iii) Matrix form of representing the transmission network.

[2]

c) Why is it important to carry out short circuit analysis for planning and operation of power systems? State four important considerations.

[4]

d) A power plant has two bus bars each connected to two generators. The two bus bars are inter-connected through a reactor. Each of the four generators is rated at 15 kV, 50 MVA with a sub-transient and transient reactance of 0.15 p.u. and 0.3 p.u., respectively. At a given point, only one of the bus bars is connected to a transmission line through a 200 MVA, 15 to 275 kV transformer with 8% leakage reactance. Calculate the value of the reactance (in p.u.) needed for the reactor between the two bus bars if the three-phase short circuit level on the generator side (i.e. low voltage side) of the transformer is to be limited to 1000 MVA. Neglect the pre-fault current and fault impedance and assume the pre-fault voltage to be 1.0 p.u. Choose 100 MVA as the system base and consider fault current contribution from the generators only.

[6]

Question 5

a) Establish the type of connection between the sequence networks for a line-to-line fault between phase B and C and show that that magnitude of the fault current (phase domain) is given by the following expression:

$$|I_f| = \frac{\sqrt{3}\,V_f}{Z_1 + Z_2 + Z_f}$$

where, Z_1, Z_2 are the equivalent positive and negative sequence impedance, Z_f is the fault impedance and V_f is the pre-fault phase voltage.

[5]

b) Determine the elements of the transformation matrix (T) which connects the phase voltages to the c-phase components of the sequence domain voltages. Assume A-B-C and A-C-B phase sequence for positive and negative sequence components, respectively.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = T \begin{bmatrix} V_{c0} \\ V_{c1} \\ V_{c2} \end{bmatrix}$$

[3]

c) A balanced three-phase supply feeds a three-phase delta connected load which draws a line current of 250 A from the supply. Due to a sudden open circuit, the current in phase A drops to zero. Under this condition, calculate the magnitude and phase angle of the positive- (A-B-C), negative- (A-C-B) and zero-sequence components of the currents in the unaffected phases using the current in Phase B as the reference (i.e. phase angle of 0 degree).

[5]

d) For a cylindrical rotor synchronous machine justify the following statements about steady-state and sub-transient values of sequence reactance with proper reasoning:

(i) The steady-state positive sequence reactance is larger than the steady-state negative sequence reactance.

[2]

(ii) Both positive and negative sequence steady-state reactance are larger than the zero sequence steady-state reactance.

[2]

(iii) Sub-transient reactance is similar for both positive and negative sequence.

[1]

e) For a three-phase transformer with delta-star connection, explain why zero sequence current can circulate within the delta connected primary winding only if the star connected winding has its neutral grounded.

[2]

Question 6

a) The power transfer capability of an uncompensated transmission line reduces with increase in distance of transmission. Draw a typical line loadability curve covering a range of transmission distance from 0 to 1000 km and explain the factors that limit the allowable power transfer for different range of distances.

[4]

b) Synchronous generators inherently provide inertial response unlike their converter-interfaced asynchronous counterparts. How is inertial response of synchronous generators helpful in reducing the drop in grid frequency?

[2]

c) The inertia constant (H) of a thermal power generator is 5 seconds. What does this statement physically signify?

[2]

d) A cylindrical rotor synchronous generator is connected to a large power system (which could be assumed to be an infinite bus) through a transmission corridor with an equivalent reactance of 0.5 p.u. The generator operates at 40% of its steady state stability limit under normal condition. The voltages at the generator terminal and infinite bus are maintained at 1.0 p.u. The resistance of the generator and the transmission corridor is negligible.

(i) The mechanical power input to the generator is increased by 20%. Use the equal area criterion to show that the maximum excursion of the generator rotor angle (δ_{max}) can be obtained from the following equation assuming a constant generator terminal voltage.

$$0.48 \times \delta_{max} + \cos \delta_{max} = 1.114$$

[6]

(ii) The power output of the synchronous generator becomes zero due to a short circuit in the transmission corridor. After the short is cleared the equivalent reactance between the generator and the infinite bus becomes 0.8 p.u. Calculate the critical clearing angle in degrees.

[6]

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